

Gavin Ford

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EDUCATION

Worcester Polytechnic Institute

Worcester, MA

BS - Electrical and Computer Engineering - GPA: 3.89

AUG 2024 - MAY 2027

- Member of Eta Kappa Nu
- Relevant coursework: RF Circuits, Embedded, Microelectronics, CMOS design.

Portland Community College

Portland, OR

AS - General Transfer Degree - GPA: 4.0

SEP 2020 - JUN 2024

- Gateway to College Scholar, 2020-2023.
- Member of Phi Theta Kappa Honor Society.

PROJECTS

Correlative Interferometry Radio Direction Finding (Phased Array)

NOV 2024

- Designed and built a CI radio direction finding system to track a rocket up to 40,000 ft. It compares the phases on five antennas using the AD8302 chip to find incident wave direction.
- Impedance controlled PCB with LNAs, filters, power splitter, phase detectors, ADCs, and an IMU.
- Used manual calculations and Advanced Design System (ADS) to design and simulate custom LNAs and 5th order bandpass filter.

Translational Drift Combat Robot (Melty Brain)

JAN 2024

- Designed, built, and programmed a competitive 150g combat robot that maximizes its moment of inertia by spinning the entire robot at 2900 RPM.
- Designed control PCB with IR beacon, accelerometer, and ESP32.
- Implemented a Kalman filter on a rotating reference frame.
- Achieved consistent orientation tracking and fast translation while spinning at 2900 RPM.

E-Paper Smartwatch

JUN 2022

- Created a functional watch that displays time and information from a connected smartphone.
- Created custom schematic and PCB layout.
- Used technologies including I2C, SPI, UART, Bluetooth, and USB.
- Developed an Android companion app to update time and weather data on the watch.
- Designed and 3D printed the watch frame.

EXPERIENCE

WPI Wireless Association — Repeater Control Operator

FEB 2025 - CURRENT

- Managed the W1WPI and W1YK radio repeater systems and setup networking for echolink nodes.
- Successfully repaired a broken cable and element on a VHF/UHF satellite station Yagi-Uda antenna.

WPI Wireless Innovation Lab — Researcher

SEP 2025 - CURRENT

- Designing an optimized testing workflow to simulate propagation on physical radio hardware.
- Developing test setup for an RF anechoic chamber and gain pattern characterizing.

WPI ECE Department — Tutor for RF Circuits Design Class

AUG 2025 - OCT 2025

- Tutored a class of 28 students on impedance matching, transmission line theory, and RF BJT/FETs.
- Held office hours to answer questions, clarify material content, and provide feedback.

Department of Conservation, New Zealand — Researcher

JAN 2025 - MAY 2025

- Worked with a team to perform in-depth research, collect surveys, and interview experts.
- Developed a comprehensive action plan for DOC to educate the public about interacting with wildlife.

Repair PDX — Small Appliance Repair Technician

JUL 2018 - AUG 2024

- Experience troubleshooting and repairing FM radios, toasters, blenders, and fans.
- Worked efficiently with others to quickly identify, debug, and repair broken devices.

SKILLS

- Equipment used: VNA, SDR, oscilloscope, network analyser, multimeter, 3D printer, CNC router.
- CAD and simulation in Cadence, Altium, KiCAD, Ansys, Matlab, ADS, Spice, Fusion 360 and OnShape.
- Micro soldering, troubleshooting, leadership, tutoring, data analysis, and documentation.
- Programming in Java, Python, C, C++, Matlab, Zig, Nix, and Bash.
- Experience with Linux, Excel, Word, PowerPoint, LaTeX, and Git.
- Amateur Extra ham radio licensed.